

Pipeline Update Report

Nanotech Media Analysis Project

April 2026

7 improvements across scraper and analysis notebook

Overview

This report summarizes 7 improvements made to the nanotech media analysis pipeline. Changes span two files: the article scraper (`create_excelV5.py`) and the analysis notebook (`nanotech_graphs.ipynb`).

What was done:

Phase 1 - Data Fixes (Scraper)

- Recovered 354 missing Science News articles
- Fixed newspaper date alignment (30% to 100% accuracy)
- Switched Year column to use reference worksheet values
- Updated date match weighting: year match = 90% minimum

Phase 2 - Visualization Fixes (Notebook)

- Removed misleading date match % for Government & Futurists
- Simplified co-occurrence graphs (top 8 keywords + "Other")

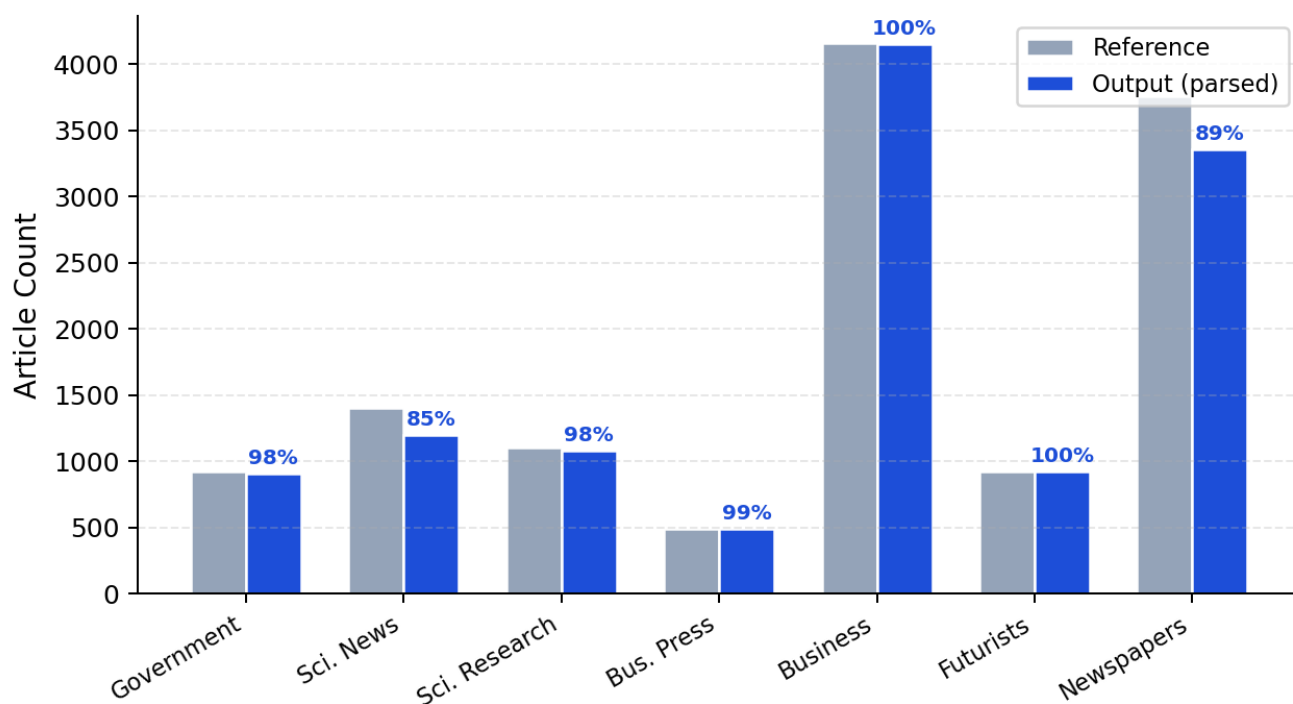
Phase 3 - Sampling Improvements (Notebook)

- Improved qualitative sampling with year-based quotas
- Added new AI keyword qualitative search sheet

Data Quality: Article Coverage

The chart below compares the number of articles in the parsed output (output.xlsx) against the reference worksheet (Combined community worksheet.xlsx). Overall coverage is 94.8% (12,106 of 12,774 articles). Futurists has perfect coverage. The main gaps are in Science News and Newspapers, where some articles in the reference could not be matched to entries in the raw text files.

Article Coverage: Output vs Reference



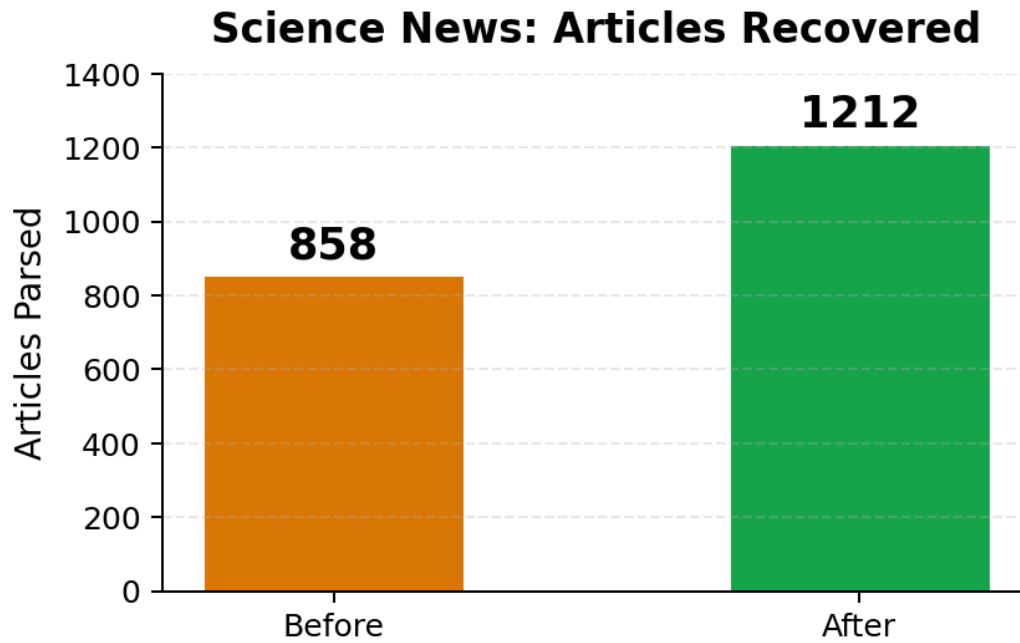
Article Count: Output vs Reference Worksheet

Source	Reference	Output	Diff	Coverage
Government	926	905	-21	97.7%
Science News	1407	1197	-210	85.1%
Sci. Research	1102	1079	-23	97.9%
Business Press	494	491	-3	99.4%
Business	4157	4154	-3	99.9%
Futurists	926	926	+0	100.0%
Newspapers	3762	3354	-408	89.2%
TOTAL	12774	12106	-668	94.8%

Phase 1: Data Fixes

1. Science News - Recovering Missing Articles

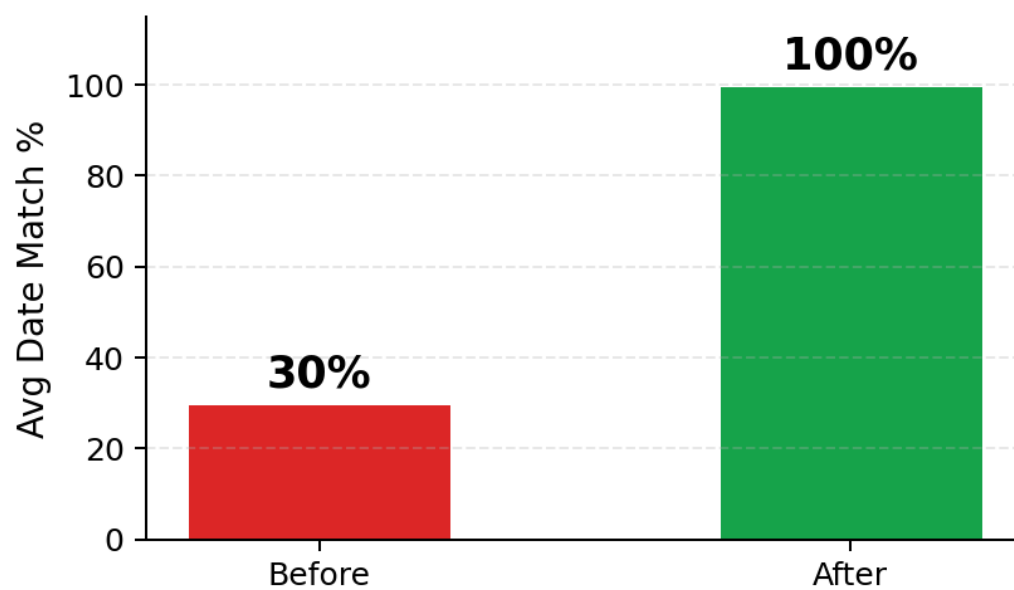
The scraper was discarding articles where the article content was incorrectly identified as the title, leaving the body empty. This mostly affected short "This Week in Science" summaries. The fix detects when a title is unusually long (>20 words) with an empty body, and recovers the content.



2. Newspaper Date Alignment

Newspaper articles were being matched to reference dates using a simple positional index (1st parsed article = 1st reference row, etc.). But the parser produced 3,354 articles while the reference had 3,762 rows, causing dates to drift out of alignment. The fix groups articles by date and matches within each date group, achieving 100% accuracy.

Newspapers: Date Alignment Accuracy



3. Year Column Now Uses Reference Worksheet

Previously, the Year column was derived from the scraped date. Now it uses the Year value from the reference worksheet (Combined community worksheet.xlsx), which is more reliable since it was manually verified. Falls back to scraped date year when no reference is available.

4. Date Match Weighting Updated

The accuracy scoring now gives heavier weight to year-level matches, since correct year is the most important factor for the temporal analysis. Scoring: exact date = 100%, same year+month = 95%, same year = 90%, wrong year = 0%. Government and Futurists are marked N/A since their source files have no inline dates.

Date Match % by Source

Source	Date Match %
Government	N/A
Science News	94.5%
Sci. Research	99.8%
Business Press	100.0%
Business	99.1%
Futurists	N/A
Newspapers	100.0%

Phase 2: Visualization Fixes

5. Gov/Futurist Date Match % Marked as N/A

The notebook was showing misleading date match percentages for Government and Futurists. These numbers were low simply because those source files lack inline dates, not because of scraper errors. They are now displayed as "N/A" while keeping the article counts visible.

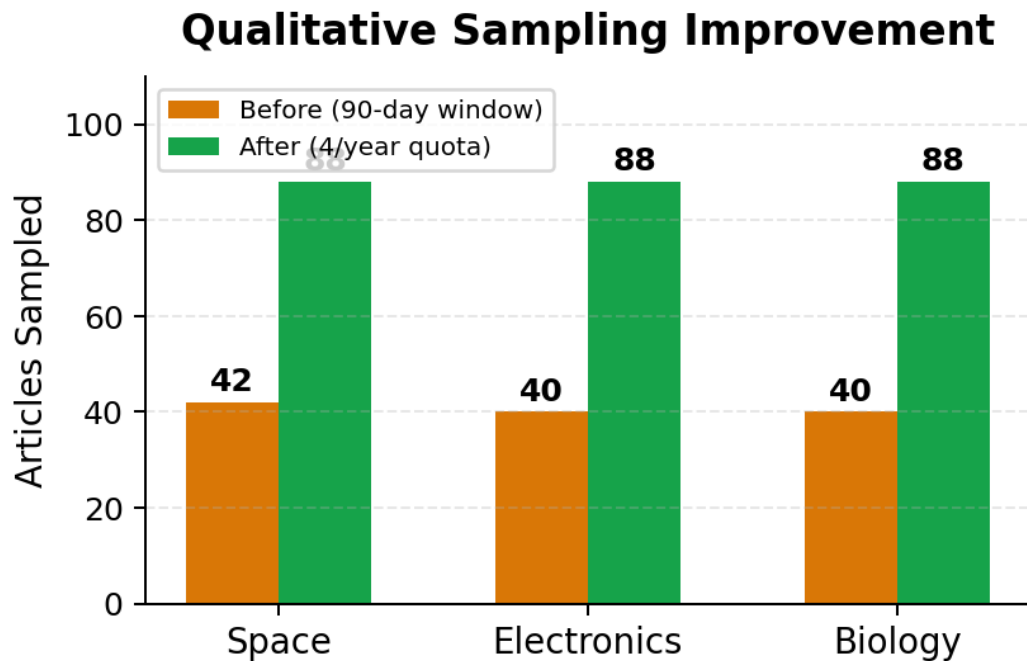
6. Co-occurrence Graphs Simplified

The co-occurrence graphs were plotting all 21 keywords per source, each with a different color. With so many overlapping lines, the graphs were unreadable. Now only the top 8 keywords (ranked by total co-occurrence) are shown with distinct colors. The remaining keywords are aggregated into a single gray "Other" line.

Phase 3: Sampling Improvements

7. Qualitative Sampling - Year-Based Quotas

The old sampling method picked one article every 90 days, which meant some years could end up with fewer than 4 samples if articles were clustered. The new method targets exactly 4 articles per year per topic. If a year has fewer than 4 qualifying articles, the shortfall carries forward to the next year, ensuring no data is lost.



8. New AI Qualitative Search Sheet

A new "AI Qualitative Samples" sheet has been added to output.xlsx. It works the same way as the existing qualitative sampling (Space, Electronics, Biology) but filters for articles mentioning both Nanotech and Artificial Intelligence. By default it samples from Futurists, but the source can be easily changed to any single source or all sources by adjusting a single variable in the notebook.

Deferred / Future Work

- Claude API analysis: Use Claude to automatically code sampled articles on qualitative dimensions (attitude toward nanotech, analogy usage, etc.). Requires setting up an Anthropic API key.
- Community dimension graphs: Once articles are coded on the qualitative dimensions, generate temporal graphs showing how different communities (Government, Business, Futurists, etc.) differ over time.